

EXPERIMENT – 03 GO FURTHER – 01

PROJECT NAME

8-Bit Arduino Function Generator using DAC and Op-Amp

OBJECTIVE / PROBLEM STATEMENT

This experiment helps in understanding waveform generation using Arduino UNO. The project generates sine, square, and triangle waves using an 8-bit DAC system with op-amp buffering and RC filtering. The experiment demonstrates concepts of Direct Digital Synthesis (DDS), Digital-to-Analog Conversion (DAC), waveform sampling, frequency control, amplitude control, and OLED interfacing.

COMPONENTS USED

Component	Quantity
Arduino UNO	1
8-Bit DAC / R-2R DAC	1
SSD1306 OLED Display	1
Op-Amp	1
10k Potentiometer	2
Push Switch	4
Resistors	Multiple
Capacitor	1
Oscilloscope	1

PIN CONNECTIONS

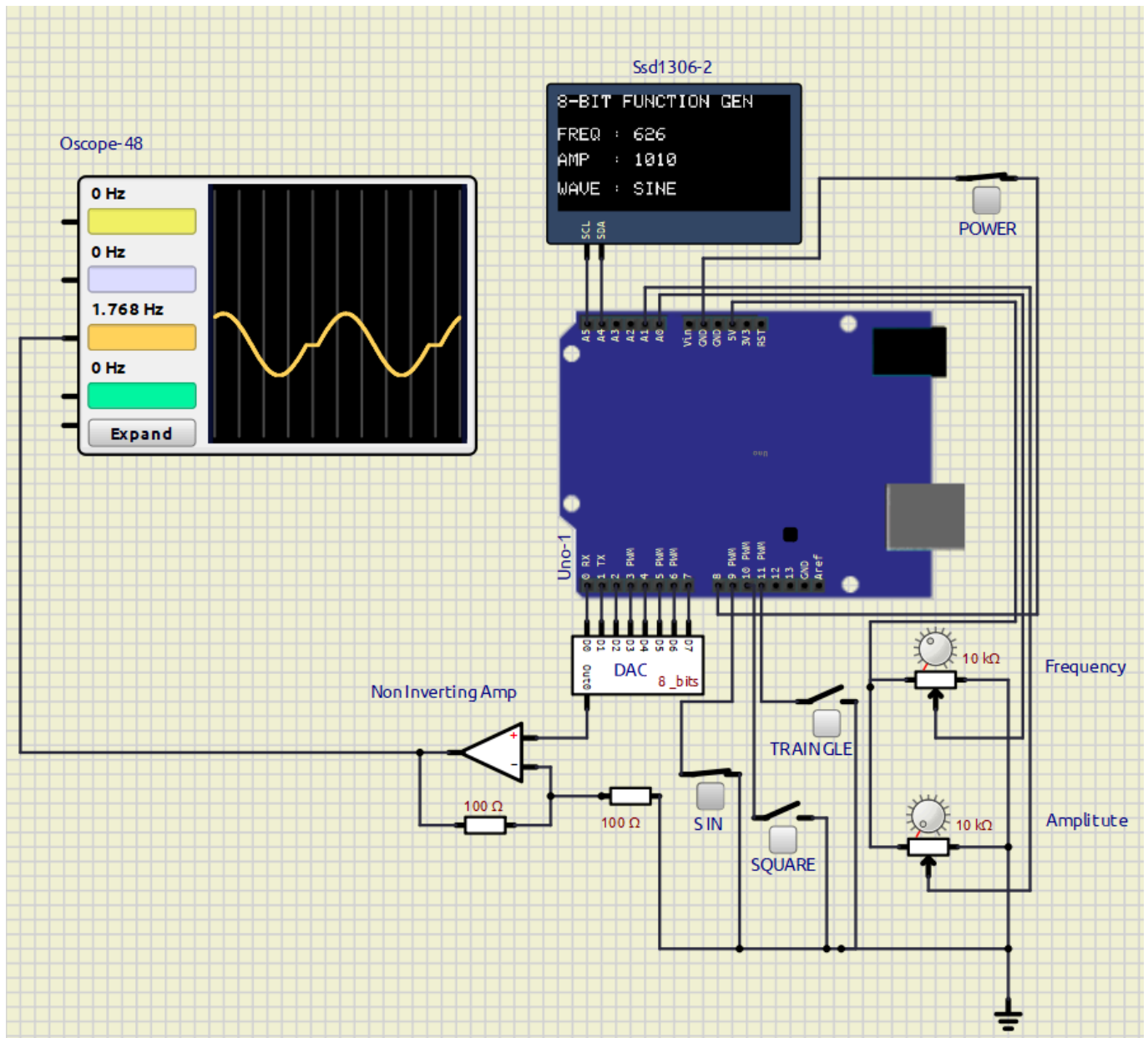
Device	Arduino Pin
DAC D0	D0
DAC D1	D1
DAC D2	D2
DAC D3	D3
DAC D4	D4
DAC D5	D5

Device	Arduino Pin
DAC D6	D6
DAC D7	D7
Power Switch	D8
Sine Wave Switch	D9
Square Wave Switch	D10
Triangle Wave Switch	D11
Frequency Potentiometer	A0
Amplitude Potentiometer	A1
OLED SDA	A4
OLED SCL	A5

SOFTWARE USED

- Arduino IDE
- SimulIDE

CIRCUIT DIAGRAM



MAIN ARDUINO CODE

```

/*
Function Generator Using UNO Board (Op-Amp + DAC 8 bit)
*/

#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <math.h>

// =====
// OLED SETTINGS

```

```
// =====

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
#define OLED_ADDR 0x3C //OLED I2C working address

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

// =====
// PIN DEFINITIONS
// =====

#define FREQ_POT      A0 //frequency potentiometer connected to A0
#define AMP_POT       A1 // Amplitude potentiometer connected to A1

#define POWER_SW      8  //System Power Botton

#define SINE_SW       9  //sine wave generate
#define SQUARE_SW     10 //square wave generate
#define TRI_SW        11 //Traingular wave generate

// =====
// VARIABLES
// =====

uint8_t sineTable[256]; //formation of sine table

int freqValue;
int ampValue;

int freqDelay;

uint8_t ampScaled;

// =====
// SETUP
// =====

void setup()
{
    // -----
    // D0-D7 AS OUTPUT
    // -----

    DDRD = 0xFF; //

    PORTD = 0;

    // -----
```

```

// SWITCHES
// -----

pinMode(POWER_SW, INPUT_PULLUP); //Assign Function to all switchs as per ther
role

pinMode(SINE_SW, INPUT_PULLUP);

pinMode(SQUARE_SW, INPUT_PULLUP);

pinMode(TRI_SW, INPUT_PULLUP);


// -----
// OLED START (Initialising Display)
// -----

display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR);

display.clearDisplay(); //clear display

display.setTextSize(1); //text size set

display.setTextColor(WHITE); //text colour set

display.setCursor(0,0); // Set initial coordinnate for displaying text

display.println("FUNCTION GENERATOR"); //Printing Function Generator

display.display(); //update display

delay(1000); //delay of 1s


// -----
// BUILD 256 SAMPLE SINE TABLE
// -----

for(int i=0; i<256; i++)
{
    float angle;

    angle = (2.0 * PI * i) / 256.0;

    sineTable[i] = (uint8_t)(127.5 + 127.5 * sin(angle));
}
}


// =====
// MAIN LOOP
// =====

```

```
void loop()
{
    // =====
    // POWER SWITCH
    // =====

    if(digitalRead(POWER_SW) == HIGH)
    {
        PORTD = 0;
        display.clearDisplay();

        display.setTextSize(2);

        display.setTextColor(WHITE);

        // "SYSTEM" centered
        display.setCursor(22,10);
        display.println("SYSTEM");

        // "OFF" centered
        display.setCursor(40,35);
        display.println("OFF");

        display.display();
        return;
    }

    // =====
    // READ POTS
    // =====

    freqValue = analogRead(FREQ_POT);

    ampValue  = analogRead(AMP_POT);

    // =====
    // FREQUENCY CONTROL
    // =====

    /*
    smaller delay = higher frequency
    */

    freqDelay = map(freqValue,0,1023,5000,60);

    // =====
    // AMPLITUDE CONTROL
    // =====

    ampScaled = map(ampValue,0,1023,0,255);
```

```
// =====  
// WAVEFORM SELECTION  
// =====  
  
if(digitalRead(SINE_SW) == LOW)  
{  
    generateSine();  
}  
  
else if(digitalRead(SQUARE_SW) == LOW)  
{  
    generateSquare();  
}  
  
else if(digitalRead(TRI_SW) == LOW)  
{  
    generateTriangle();  
}  
  
else  
{  
    PORTD = 0;  
}  
  
// =====  
// OLED UPDATE  
// =====  
  
updateDisplay();  
}  
  
// =====  
// AMPLITUDE SCALE FUNCTION  
// =====  
  
uint8_t scaleAmplitude(uint8_t value)  
{  
    return ((unsigned int)value * ampScaled) / 255;  
}  
  
// =====  
// SINE WAVE  
// =====  
  
void generateSine()  
{  
    for(int i=0; i<256; i++)  
    {
```

```
        PORTD = scaleAmplitude(sineTable[i]);

        delayMicroseconds(freqDelay);
    }
}

// =====
// TRIANGLE WAVE
// =====

void generateTriangle()
{
    for(int i=0; i<256; i++)
    {
        PORTD = scaleAmplitude(i);

        delayMicroseconds(freqDelay);
    }

    for(int i=255; i>=0; i--)
    {
        PORTD = scaleAmplitude(i);

        delayMicroseconds(freqDelay);
    }
}

// =====
// SQUARE WAVE - FIXED VERSION
// =====

void generateSquare()
{
    uint8_t highLevel;

    highLevel = scaleAmplitude(255);

    // FIXED: High level first, then low level
    PORTD = highLevel;

    delayMicroseconds(freqDelay * 128);

    PORTD = 0;

    delayMicroseconds(freqDelay * 128);
}

// =====
```



```
// OLED DISPLAY
// =====

void updateDisplay()
{
    static unsigned long lastUpdate = 0;

    if(millis() - lastUpdate < 150)
        return;

    lastUpdate = millis();

    display.clearDisplay();

    display.setTextSize(1);

    display.setCursor(0,0);

    display.println("8-BIT FUNCTION GEN");

    display.setCursor(0,18);

    display.print("FREQ : ");

    display.println(freqValue);

    display.setCursor(0,32);

    display.print("AMP  : ");

    display.println(ampValue);

    display.setCursor(0,48);

    display.print("WAVE : ");

    if(digitalRead(SINE_SW) == LOW)
    {
        display.println("SINE");
    }

    else if(digitalRead(SQUARE_SW) == LOW)
    {
        display.println("SQUARE");
    }

    else if(digitalRead(TRI_SW) == LOW)
    {
        display.println("TRIANGLE");
    }
}
```

```
}  
  
else  
{  
    display.println("NONE");  
}  
  
display.display();  
}
```

WORKING PRINCIPLE

1. Arduino reads frequency and amplitude values using potentiometers.
2. Waveform type is selected using switches.
3. A sine lookup table stores 256 waveform samples.
4. Arduino sends digital data through PORTD (D0–D7).
5. The DAC converts digital values into analog voltage.
6. The op-amp buffers and stabilizes the waveform.
7. The RC filter smooths the staircase output.
8. The oscilloscope displays the generated waveform.

WHAT I LEARN

- Understanding of DAC and DDS concepts
- Fast digital output using PORTD
- Frequency control using timing delay
- Waveform generation using lookup tables
- OLED interfacing with Arduino

APPLICATIONS

- Signal testing
- Audio waveform experiments
- Electronics laboratory
- Communication system experiments
- Embedded systems learning
- DAC and waveform analysis

CONCLUSION

The function generator was successfully designed and simulated using Arduino UNO, DAC, op-amp, and RC filter. The project demonstrated waveform generation using Direct Digital Synthesis (DDS) and helped in understanding DAC operation, waveform sampling, signal conditioning, and embedded system programming.

REFERENCES

1. Arduino Official Website
<https://www.arduino.cc/>
2. SSD1306 OLED Documentation
https://github.com/adafruit/Adafruit_SSD1306
3. Adafruit GFX Library
<https://github.com/adafruit/Adafruit-GFX-Library>
4. SimulIDE Official Website
<https://simulide.com/>